VARUNA 2047 — ForestWatch Intelligence Report

Region: WAR | **Type:** District | **Generated:** 2026-10-27 06:21 UTC | **Classification:** Research Grade

1. EXECUTIVE SUMMARY & CANOPY INTEGRITY

This evidence dossier presents monitored canopy changes and satellite observations for WAR. Under the Forest Survey of India (ISFR) benchmark, baseline forest extent is recorded at **38,200 hectares**. During the observation window, measured tree-cover loss area totaled **42.5 hectares** (0.111% of baseline).

2. TREE-LOSS METRICS & DEFENSIBLE ESTIMATION

Metric Indicator	Measured Value	Scientific Standard
Measured Tree-Cover Loss Area	42.5 ha	Hansen GFC v1.11 / Sentinel-2 dNDVI (10m)
Percentage Canopy Change	0.111%	100 * (Loss Area / Baseline Forest Area)
Estimated Tree Stems Lost	N/A	FSI NFI Average Stem Density per Hectare (+/- 25%)
Change Trajectory	STABLE	Forest loss rate remains within historical baseline bounds (+/- 15%
Baseline Sentinel-2 Scene	S2A_MSIL2A_20190114T050151_N0211_R0...	ESA Copernicus BOA Surface Reflectance
Recent Sentinel-2 Scene	S2B_MSIL2A_20240118T050149_N0510_R0...	ESA Copernicus BOA Surface Reflectance

3. SCIENTIFIC TRANSPARENCY & LIMITATION DISCLOSURES

Important Methodological Note: Tree-cover loss detected: 42.5 hectares. Individual trees lost cannot be reliably determined from the available satellite resolution.

Spaceborne multispectral imagery (10m–30m) captures canopy-cover disturbance rather than individual tree stems. Tree counts cannot be defensibly calculated without sub-meter crown-detection sensors or field inventory plots. Canopy-loss pixels do NOT imply clearcut logging or illegal felling without ground enforcement records.